INCH-POUND
MIL-DTL-24299B(SH)
17 October 2017
SUPERSEDING
MIL-H-24299A(SH)
5 August 1994

DETAIL SPECIFICATION

HEATERS, LUBRICATING OIL AND FUEL, ELECTRIC, NAVAL SHIPBOARD

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers naval shipboard heaters that are stationary, self-contained, and electrically operated, which are controlled and monitored by an external supervisory network or a human watchstander. The heaters are used to heat lubricating oil or fuel for purification or warm-up.

1.2 Classification. Heaters are of the following types and styles as specified (see 6.2):

a. Type A – Immersion heater
b. Type B – Circulation heater
   (1) Style 1 – Horizontal mount
   (2) Style 2 – Vertical mount

1.3 Part or identifying number (PIN). PINs to be used for heaters acquired to this specification are created as follows:

M	**24299**	**-**	**X**	**-**	**X**
Prefix for Military Specification	Specification Number		Type (A or B)		Style (Type B only) (1 or 2)
Examples: M24299-A, M24299-B-2					

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4540

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-286	-	Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)
TT-P-645	-	Paint, Alkyd Type, Formula Number 84

FEDERAL STANDARDS

FED-STD-791	-	Testing Method of Lubricants, Liquid Fuels, and Related Products

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-917	-	Electric Power Equipment, Basic Requirements for
MIL-PRF-1149	-	Gasket Materials, Synthetic Rubber, 50 and 65 Durometer Hardness
MIL-DTL-1222	-	Studs, Bolts, Screws and Nuts for Applications Where a High Degree of Reliability is Required; General Specification for
MIL-DTL-2036	-	Enclosures for Electric and Electronic Equipment, Naval Shipboard
MIL-DTL-2212	-	Contactors and Controllers, Electric Motor AC or DC, and Associated Switching Devices
MIL-DTL-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-PRF-9000	-	Lubricating Oil, Shipboard Internal Combustion Engine, High-Output Diesel
MIL-DTL-15024	-	Plates, Tags, and Bands for Identification of Equipment, General Specification for
MIL-DTL-15090	-	Enamel, Equipment, Light Gray (Navy Formula No. 111)
MIL-C-15726	-	Copper-Nickel Alloy, Sheet, Plate, Strip, Bar, Rod, and Wire
MIL-T-16420	-	Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706)
MIL-DTL-16884	-	Fuel, Naval Distillate
MIL-PRF-17331	-	Lubricating Oil, Steam Turbine and Gear, Moderate Service
MIL-PRF-20042	-	Flanges, Pipe, Bronze (Silver Brazing)
MIL-H-22577	-	Heating Elements Electrical: Cartridge, Strip and Tubular Type
MIL-S-22698	-	Steel Plate, Shapes and Bars, Weldable Ordinary Strength and Higher Strength: Structural
MIL-PRF-23648	-	Resistor, Thermal (Thermistor) Insulated, General Specification for
MIL-PRF-23699	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Numbers: O-152, O-154, O-156, and O-167

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MIL-T-24388 - Thermocouple and Resistance Temperature Detector Assemblies, General Specification for (Naval Shipboard)

MIL-P-24691 - Pipe and Tube, Carbon, Alloy and Stainless Steel, Seamless and Welded, General Specification for

MIL-P-24691/1 - Pipe and Tube, Carbon Steel, Seamless

MIL-G-24716 - Gaskets, Metallic-Flexible Graphite, Spiral Wound

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

MIL-STD-202 - Electronic and Electrical Component Parts

MIL-STD-769 - Insulation Requirements for U.S. Naval Vessels

MIL-STD-1399-300 - Electric Power, Alternating Current

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment

MIL-HDBK-454 - General Guidelines for Electronic Equipment

MIL-HDBK-470 - Designing and Developing Maintainable Products and Systems, Volume I

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMITTEE ON NATIONAL SECURITY SYSTEMS INSTRUCTION (CNSSI)

CNSSI No. 1253 - Security Categorization and Control Selection for National Security Systems

(Copies of this document are available online at https://www.cnss.gov/CNSS/issuances/Instructions.cfm.)

MILITARY SPECIFICATION SHEETS

MS18229 - Plug for O-Ring Gasket

(Copies of this document are available online at http://quicksearch.dla.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NASM15795	-	Washer, Flat, Metal, Round, General Purpose
NASM16187	-	Stud, Continuous Thread, Bolt-Stud, B16 Alloy Steel, High Temperature Service 875-1,000 °F
NASM17828	-	Nut, Self-Locking, Hexagon Regular-Height, (Non-Metallic Insert) 250 °F or 450 °F, Nickel-Copper Alloy
NASM17829	-	Nut, Self-Locking, Hexagon, Regular Height, 250 °F, Non-Metallic Insert, Non-CRES Steel
NASM17830	-	Nut, Self-Locking, Hexagon-Regular, 250 °F and 450 °F, Non-Metallic Insert, 300 Series CRES
NASM21044	-	Nut, Self-Locking, Hexagon, Regular Height, 250 °F, 125 KSI Ftu and 60 KSI Ftu
NASM35307	-	Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Corrosion Resisting, Passivated, UNC-2A
NASM35649	-	Nut, Plain, Hexagon, Machine Screw, UNC-2B

(Copies of these documents are available online at www.aia-aerospace.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5	-	Pipe Flanges and Flanged Fittings: NPS ½ Through NPS 24 Metric/Inch Standard

(Copies of this document are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A182/A182M	-	Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service
ASTM A204/A204M	-	Standard Specification for Pressure Vessel Plates, Alloy Steel, Molybdenum
ASTM A240/A240M	-	Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM A312/A312M	-	Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes
ASTM A515/A515M	-	Standard Specification for Pressure Vessel Plates, Carbon Steel, for Intermediate- and Higher-Temperature Service
ASTM A516/A516M	-	Standard Specification for Pressure Vessel Plates, Carbon Steel, for Moderate- and Lower-Temperature Service
ASTM D974	-	Standard Test Method for Acid and Base Number by Color-Indicator Titration
ASTM E230/E230M	-	Standard Specification and Temperature-Electromotive Force (emf) Tables for Standardized Thermocouples

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ASTM F468	-	Standard Specification for Nonferrous Bolts, Hex Cap Screws, Socket Head Cap Screws, and Studs for General Use
ASTM F1836M	-	Standard Specification for Stuffing Tubes, Nylon, and Packing Assemblies (Metric)
ASTM F2329/F2329M	-	Standard Specification for Zinc Coating, Hot-Dip, Requirements for Application to Carbon and Alloy Steel Bolts, Screws, Washers, Nuts, and Special Threaded Fasteners

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AMS 3216	-	Fluorocarbon (FKM) Rubber, High-Temperature - Fluid Resistant, Low Compression, Set 70 to 80
SAE AMS 3218	-	Fluorocarbon (FKM) Rubber, High-Temperature-Fluid Resistant, Low Compression, Set 85 to 95
SAE J1926-1	-	Connections for General Use and Fluid Power – Ports and Stud Ends with ASME B1.1 Threads and O-Ring Sealing – Part 1: Threaded Port with O-Ring Seal in Truncated Housing

(Copies of these documents are available online at www.sae.org.)

UNDERWRITERS LABORATORIES, INC. (UL)

UL 574	-	Standard for Electric Oil Heaters
UL 823	-	Standard for Electric Heaters for Use in Hazardous (Classified) Locations

(Copies of these documents are available online at http://www.ul.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Heaters furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.1.1 Operating fluid. The heater will be used to heat lubricating oil or fuel as specified (see 6.2).

3.2 Materials. Materials shall be as specified in table I, table II, or table III (see 6.2). In cases where a choice of materials is given in the tables and a specific material is preferred, the acquisition requirements shall so indicate.

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TABLE I. Ferrous materials.

Part	Material	Applicable document
Shell and pipe connections	Tubing, seamless drawn carbon steel	MIL-P-24691
		MIL-P-24691/1
	Plate, welded, carbon-silicon steel, marine boiler quality	ASTM A515/A515M
		ASTM A516/A516M
		ASTM A204/A204M
Head and mounting plate	Plate, carbon-silicon steel, marine boiler quality	ASTM A515/A515M
		ASTM A516/A516M
		ASTM A204/A204M
Flanges for shell and pipe connections	Steel, carbon	ASME B16.5
Plates and mounting lug	Plate, carbon-silicon steel, marine boiler quality	ASTM A515/A515M
		ASTM A516/A516M
		ASTM A204/A204M
	Plate, steel, carbon, structural	MIL-S-22698
Stay rods	Steel, carbon	MIL-S-22698
Plugs	Steel, carbon	MS18229
Studs	Steel (grade B16)	NASM16187
Screws, bolts	Steel (grade 5 or grade B16)	MIL-DTL-1222
Nuts	Steel, carbon, zinc coated	ASTM F2329/F2329M
Nuts, self-locking	Steel, zinc plated	NASM17829
		NASM21044
Washers	Steel, carbon, zinc coated	ASTM F2329/F2329M

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TABLE II. Nonferrous materials.

Part	Material	Applicable document
Shell and pipe connections	Copper-nickel alloy, composition 90-10	MIL-T-16420
	Copper-nickel alloy, composition 70-30	MIL-C-15726
Head and mounting plate	Copper-nickel alloy	MIL-C-15726
Flanges for shell and pipe connections 1/	Copper-nickel alloy	MIL-C-15726
Plates and mounting lugs	Copper-nickel alloy	MIL-C-15726
Stay rods	Nickel-copper-aluminum alloy	QQ-N-286
	Copper-nickel alloy, composition 70-30	MIL-C-15726
Studs	Nickel-copper	MIL-DTL-1222
	Nickel-copper-aluminum	
Screws, bolts	Nickel-copper	ASTM F468
	Nickel-copper-aluminum	MIL-DTL-1222
Nuts	Nickel-copper	NASM35649
Nuts, self-locking	Nickel-copper	NASM17828
Washers	Nickel-copper	NASM15795
Plugs	Nickel-copper alloy	MS18229
	Copper-nickel alloy	

NOTE:
1/ Nonferrous flanges shall meet the dimensional requirements in accordance with MIL-PRF-20042.

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TABLE III. Corrosion resistant (CRES) materials.

Part	Material	Applicable document
Shell and pipe connections	304L or 316L	ASTM A312/A312M
Head and mounting plate	304L or 316L	ASTM A240/A240M
Flanges for shell and heater and pipe connections	304L or 316L	ASTM A182/A182M
Plates and mounting supports	Plate, carbon-silicon steel, marine boiler quality	ASTM A515/A515M
		ASTM A516/A516M
		ASTM A204/A204M
	Plate, steel carbon structural	MIL-S-22698
Stay rods	CRES	ASTM A312/A312M
Plugs	CRES	MS18229
Screws, bolts	CRES	NASM35307
	Corrosion and Heat Resistant Steel (CHRES) (A286 Alloy)	MIL-DTL-1222
Nuts	CRES	NASM35649
	CHRES (A286 Alloy)	MIL-DTL-1222
Nuts, self-locking	CRES	NASM17830

3.2.1 Gaskets and stuffing tubes. The following materials shall be used for gaskets and stuffing tubes, as applicable:

a. Metallic Flexible Graphite, Spiral Wound in accordance with MIL-G-24716.

b. O-rings in accordance with SAE AMS 3216 (SAE AMS 3218, 90 durometer for threaded bosses).

c. Synthetic rubber in accordance with MIL-PRF-1149.

d. Stuffing tubes, nylon in accordance with ASTM F1836M.

3.2.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.3 Prohibited materials. Asbestos, cadmium, and mercury shall not be used in the manufacture of the heaters or controls. In addition, zinc plating of parts to be immersed in oil is prohibited.

3.3 Construction. Construction shall be as specified in 3.3.1 through 3.3.15.9 and UL 823.

3.3.1 Heating unit assembly. The heating unit assembly shall consist of a circular mounting flange through which heater elements are inserted and welded to the plate at each penetration. The other end of the elements shall have spacers, or be braced between elements, for rigidity and shall have a means of support to the tank or heater shell. The unit shall be constructed to permit easy removal of connected piping. Mounting flanges shall have threaded breakaway holes. These holes may also be used for inserting lifting eyebolts. All threaded fasteners to be immersed in the heated fluid shall be welded to prevent loosening.

3.3.2 Heater elements. Heater elements shall be type III tubular in accordance with MIL-H-22577 with seamless sheathing and ceramic-to-metal hermetic type end seals. Unless otherwise specified (see 6.2), sheath material shall be as specified in table IV. Heating elements shall be bright annealed after all fabrication steps up to insertion into the mounting plate.

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TABLE IV. Sheath material.

Fluid	Type of heater	Sheath material
Lube oil	A	CRES 304L or 316L
	B	Steel or 316L
Fuel, service	A or B	CRES 304L or 316L
Fuel, transfer	B	Inconel 625

3.3.3 Terminal box and control enclosures. The electrical connections to the heater elements shall be enclosed in a watertight terminal box mounted on a flange and provided with a removable cover, selected and designed in accordance with MIL-DTL-2036. The terminal box and other electrical and electronic enclosures shall be lockable as specified (see 6.2). These enclosures shall have no externally accessible data ports. A connection wiring diagram shall be glued to the underside of the terminal box cover. Holes for cable connections to the terminal box shall be provided as specified (see 6.2). Terminal connections, electrical creepage, and clearance distances shall be in accordance with MIL-DTL-917. Further guidance may be found in MIL-HDBK-454.

3.3.4 Mounting flange. The mounting flange shall be marked to ensure installation of the heating unit assembly in only one position. In order to ensure interchangeability, the tolerance for the location of the drilled flange holes shall be within 0.010 inch of the true design location. The mounting surface shall be finished to ensure a leak-tight seal with the gasket used (see 3.3.14.1.a and 6.2).

3.3.5 Temperature sensing devices. Temperature sensing devices shall be provided with each heater and shall be of the types specified (see 6.2). Unless otherwise specified (see 6.2), the same type of sensor used to input to the controller shall be used to input to alarms, controls, and protective devices. These sensors shall have no externally accessible data ports, and their programmable parameters limited to within ±25 percent of the specified operating range (see 6.2). Temperatures shall be monitored and reported to the controller as required (see 6.2). Sensors need not have onboard memory but shall be capable of sending temperature readings to an external supervisory network. The external supervisory network will log, time stamp, and protect this data from unauthorized access, modification, and deletion.

3.3.5.1 Temperature sensing device operation. Temperature sensing device operation shall be verified when heaters are tested for conformance as specified in 4.3.

3.3.5.2 Thermostatic switch. Thermostatic operating switches shall be of an adjustable immersed liquid filled bulb type in accordance with MIL-DTL-2212. Thermostatic switch installation shall allow for periodic removal for calibration or replacement without damage.

3.3.5.3 Thermocouples. Thermocouples shall be type K or type J in accordance with ASTM E230/E230M. At a minimum, a thermocouple shall be installed 2 inches from and centered on the outlet of the heater. Thermocouple connections shall be designed such that they cannot be reverse wired, so as to prohibit the possibility of the unit to be constantly energized.

3.3.5.4 Thermistor. When specified (see 6.2), a thermistor temperature detector in accordance with MIL-PRF-23648 shall be used in conjunction with a solid-state control device. The thermistor shall have no externally accessible data ports, and their programmable parameters shall be limited to within ±25 percent of the specified operating range. In the event of a failure, they shall fail to a safe default state, generating an alert to the supervisory network.

3.3.5.5 Resistance temperature element (RTE). When specified (see 6.2), an RTE in accordance with MIL-T-24388 shall be used in conjunction with a solid-state control device. The RTE shall have no externally accessible data ports, and their programmable parameters shall be limited to within ±25 percent of the specified operating range. In the event of a failure, they shall fail to a safe default state, generating an alert to the supervisory network.

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3.3.6 Heaters.

3.3.6.1 Type A heaters. Applications using type A heaters generally require a heat-up time of approximately 1 hour during which the heaters are energized 100 percent of the time. Immersion heaters are generally installed in tanks. When the heater is on, heat transfer in tanks occurs by conduction and convection at a rate of rise of about 1 degree per minute. Controllers for immersion heaters are generally of the magnetic type. Controllers may be designed to heat up to a specified temperature and shut off, or they may control around a specific temperature.

3.3.6.2 Type B heaters. Applications using type B heaters generally require 50 to 120 °F temperature rise in the period of time (5 seconds or less) that it takes for one pass through the heater. If the outlet of the heater flows into a tank, control of the tank temperature is more appropriate than control of the heater outlet temperature.

3.3.6.3 Heater controls. Control of tank temperature may require magnetic controllers (see 3.3.6.3.a) or a stepper control (see 3.3.6.3.b). If the outlet of the heater does not flow into a tank, solid-state controllers (see 3.3.6.3.c) and proportional integral derivative (PID) controllers (see 3.3.6.3.d) may be required to maintain outlet temperature. Lubricating oil heaters shall require additional controls to preclude localized overheating of the oil (see 3.3.6.3.g). In all cases, heater controls shall have cybersecurity measures (see 3.3.6.3.h) implemented. Password protection at the individual heater shall not be required. The supervisory network controlling the heater shall have such protections, but is beyond the scope of this specification.

a. Magnetic controller. When specified (see 6.2), a magnetic contactor shall be furnished with each heater. The contactor shall be in accordance with MIL-DTL-2212. The heater control device shall function to de-energize the heater within ±10 °F of the adjusted temperature setting. At the time the control device functions, the maximum temperature of the fluid in any part of the heater shall not exceed the adjusted temperature setting by greater than 15 °F. The heater shall be re-energized before the temperature has dropped to 90 percent of the adjusted temperature setting.

b. Stepper control. When specified (see 6.2), a circular stepper power control shall be furnished for each heater. Stepper power control shall be by a magnetic across-the-line type controller with the heater load divided into equal parts to be sequentially energized via contactors with a 10-second delay between each sequence and de-energized in a first-on-first-off order. Sequencing control shall be by solid-state control.

c. Solid-state controller. When specified (see 6.2), a solid-state controller shall be furnished. The heater control device in this instance shall de-energize the heater within ±5 °F of the adjusted temperature setting. At the time the control device functions, the maximum temperature of the fluid in any part of the heater shall not exceed the adjusted temperature setting by more than 8 °F. The heater shall be re-energized before the temperature of the fluid has dropped to 95 percent of the adjusted temperature setting. The controller shall cycle the heating elements by a solid-state thyristor power controller to maintain the specified heater outlet temperature. The solid-state thyristor control shall provide enforced zero firing power control. Heater elements shall be divided into three or more circuits having equal kilowatt ratings. Under steady-state conditions, individual circuits shall be brought from 0 to 100 percent of the rating sequentially with increasing demand for heat. The cycle period for thyristor control shall be field adjustable. Thyristors shall be protected by fuses of the type intended for protecting semi-conductor devices. Thyristor heat sinks shall be of designs capable of normal operation under the ambient temperatures encountered. The controllers shall shut down the heater upon loss of any thermocouple signal.

d. PID controller. When specified (see 6.2), a PID controller shall be furnished with each heater. PID controllers shall be implemented in programmable logic controllers (PLCs) or as a panel-mounted digital controller. Heater control temperature shall be sensed by thermocouple. Heater controllers shall be of the magnetic controller/solid-state thyristor type. The controller shall be in accordance with MIL-DTL-2212 and the controller enclosure shall contain the solid-state PID temperature controller, element over-temperature control, magnetic contactors for shorted silicon controlled rectifier (SCR), and a solid-state SCR thyristor device for proportioning power to heating elements and control transformers. There shall be a port installed on the controller such that operation of the SCR displays can be viewed without opening the controller door.

e. Power requirements (see 6.2).

f. Specific degree of heater control required (see 6.2).

g. Lubricating oil heaters shall, at a minimum, be provided three digital controllers consisting of a process/outlet controller, a high sheath temperature controller, and a high limit controller.

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h. All controls and their associated circuitry and enclosures shall be in accordance with the cybersecurity measures of CNSSI No. 1253 and the following. Additional measures shall be as specified (see 6.2).

(1) There shall be no externally accessible data ports. All unused or unnecessary ports inside enclosures shall be disabled physically or logically. Physical access to the ports shall require opening an enclosure.

(2) Programmable parameters shall be limited to within ±25 percent of the specified operating range. Any changes to specified controller parameters, logging, or programs exceeding this range shall generate an alert to the supervisory network.

(3) Controls and sensors shall not be reprogrammable over the connected supervisory network but only via a direct physical port connection with password protection capability inside the control or circuitry enclosure.

(4) Neither remote nor wireless reprogramming shall be permitted.

(5) Enclosures shall be provided with fasteners, allowing use of lead-wire or other tamper-evident seals (see 3.3.3 and 6.2).

(6) Controls shall include monitoring of sensors for failures, which shall generate alerts locally, if so equipped, and to the supervisory network.

(7) Controls shall have heater default parameters and programs stored in read-only memory.

(8) The ability to create, enable, modify, disable, and remove user accounts shall be provided. Controllers should be able to support user accounts based on roles and responsibilities.

(a) Each account shall have rights in accordance with the principle of least privilege.

(b) Factory default authentication credentials shall be removed or changed prior to delivery or installation.

(9) Procedures to restore the controller and its associated sensors shall be documented and provided to Naval Sea Systems Command (NAVSEA).

(10) Baseline configurations shall be documented as specified (see 6.2). Baseline configurations include, but are not limited to, information about components (e.g., software packages, version numbers and patch information on operating systems and application, and configuration setting and parameters), network topology, and the logical placement of the components.

3.3.7 Reliability and maintainability. The heater assembly, temperature control device, and heater control device shall be designed for a mean time between failure of 100,000 or more hours of operation, where failure is defined as any condition that prevents the heater from meeting the requirements of 3.3.6.1 or 3.3.6.2, as applicable. The mean time to repair shall be no greater than 3.0 hours, and the maximum time to repair shall be no greater than 6.0 hours. The mean maintenance time per month shall be no greater than 3.0 hours.

3.3.8 Finish.

3.3.8.1 External surfaces. Unprotected external surfaces of non-corrosion resistant ferrous parts shall be treated as follows:

a. Sandblasted clean.

b. Primed with one coat of primer in accordance with TT-P-645.

c. Painted with one coat of enamel in accordance with MIL-DTL-15090.

3.3.8.2 Internal surfaces. Internal surfaces that will come in contact with the heated fluid shall not be painted, plasticized, or zinc coated.

3.3.9 Fasteners. Fasteners shall be identified by their part or identification numbers from the part standards as specified in table I, table II, and table III.

3.3.9.1 Commercial utility hardware. Commercial items such as screws, nuts, and washers may be used in non-critical applications provided they possess properties similar to and are replaceable by standard parts specified in 3.3.9.

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3.3.10 Welding and allied processes. Welding and allied processes shall be in accordance with S9074-AR-GIB-010/278.

3.3.11 Voltage. The operating voltage shall be one of the following: 440 volts alternating current (VAC), 115 VAC, special AC voltages, 115 volts direct current (VDC), or 230 VDC, as specified (see 6.2). For AC voltages including special AC voltages, equipment shall be 60 hertz and comply with MIL-STD-1399-300. For DC voltages, the element shall operate satisfactorily on voltages up to 105 percent of the specified rating. When heating elements are intended for operation on a voltage other than line voltage (by series or wye connections), the line-to-line voltage shall also be indicated. The heating element shall be insulated for the line voltage and shall have electrical creepage and clearance dimensions for line voltage in accordance with MIL-DTL-917. Unless otherwise specified (see 6.2), control devices shall operate on the same voltage as the heating elements.

3.3.12 Safety. Equipment shall be constructed to ensure safety to operating and maintenance personnel in accordance with MIL-DTL-917.

3.3.13 Over-temperature safety devices.

3.3.13.1 Element temperature. When specified (see 6.2), a temperature sensing device shall be firmly clamped to an inner heating element to measure sheath temperature and shall be set as specified in table V.

TABLE V. Over-temperature safety device setpoints.

Fluid	Applicable document	Setpoint
Lubricating oil, steam turbine (2190 TEP)	MIL-PRF-17331	400±50 °F
Lubricating oil, gas turbine (23699)	MIL-PRF-23699	400±50 °F
Lubricating oil, diesel (9250)	MIL-PRF-9000	400±50 °F
Fuel, naval distillate:		
(NATO Code F-44) (JP-5)	MIL-DTL-5624	265±50 °F
(NATO Code F-76) (DFM [1/])	MIL-DTL-16884	
NOTE: [1/] Diesel fuel marine		

3.3.13.2 Fluid temperature. When specified (see 6.2), a temperature sensor shall be installed to sense the highest local bulk fluid temperature produced by the heater. The sensor shall be internal to a type B heater. The control shall be set to interrupt power to the heater when the fluid temperature rises above the specified thermostat setpoint (see table V) and reapply power when the fluid temperature drops below the setpoint. This control shall operate automatically and shall not require manual reset nor be connected to any alarms.

3.3.14 Type A heaters. Type A heaters shall be as specified in 3.3.14.1 and 3.3.14.2.

3.3.14.1 Heater unit. The heaters shall be of the flanged immersion type with the following characteristics as specified (see 6.2):

a. Size, type, and surface finish of mounting flange.

b. Power rating.

c. Immersion length.

d. Hole diameter for heater insertion.

e. Weight limitation.

3.3.14.2 Watt density. The watt density for type A heaters shall be not greater than 16 watts per square inch of heating element.

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3.3.15 Type B heaters. Type B heaters shall be as specified in 3.3.15.1 through 3.3.15.9.

3.3.15.1 Shell. The heater assembly shall be enclosed in a shell with inlet and outlet piping connections to a lubricating oil or fuel circulation system. Style 1 (horizontal mount) heaters shall have inlet and outlet connections in an upward orientation. Style 2 (vertical mount) heaters shall have the inlet connection at the bottom of the shell and the outlet at the top.

3.3.15.2 Pipe connections. Flanged inlet and outlet connections shall be welded to the shell. Straight threaded connections in accordance with SAE J1926-1 shall be provided for draining the heater and for continuous venting. These connections shall be closed by O-ring gasketed plugs in accordance with MS18229. Pipe connections shall be of the size, type, and (for flanges) surface finish specified (see 6.2).

3.3.15.3 Heater unit. The heater unit shall be constructed so that the flow of lubricating oil or fuel over all heating surfaces is evenly distributed to prevent localized hot spots. Hot spots for the purposes of this specification are areas where fluid temperatures may exceed outlet temperatures by as much as 40 °F. This requirement applies to both vertically and horizontally mounted heaters. Heating elements shall have an unheated length which extends, as a minimum, from the flange to the centerline of the outlet nozzle.

3.3.15.4 Mounting. Heaters may have mounting lugs welded to the heater shell. The lugs shall be drilled and tapped for bolting the heater to the ship structure. The lugs shall be flush with the insulation and lagging, if provided. Clamp bands or other means of mounting to the ship's structure may also be acceptable. Specific mounting configuration shall be as specified (see 6.2).

3.3.15.5 Thermal insulation. Heaters shall be provided with a heat shield or they shall be insulated and lagged in accordance with MIL-STD-769 as specified (see 6.2). The lagging shall not interfere with removal of the heater from the shell. Heater lagging shall be jacketed; jacket material shall be CRES in accordance with ASTM A240/A240M, type T304.

3.3.15.6 Watt density. The watt density for type B heaters shall be not greater than 20 watts per square inch of heating element.

3.3.15.7 Capacity. Flow rate, inlet temperature range, outlet temperature, and required minimum temperature rise for the heated fluid shall be as specified (see 6.2).

3.3.15.8 Pressure drop. Unless otherwise specified (see 6.2), when operating at maximum rated flow and rated input power and outlet temperature, pressure drop through the heater shall be not greater than 10 pounds per square inch (lbf/in^2).

3.3.15.9 Limiting dimensions. Length, width, height, weight, and element withdrawal dimensions shall be as specified (see 6.2).

3.4 Performance characteristics. Performance characteristics shall be as specified in 3.4.1 through 3.4.6.

3.4.1 Operating pressure. Unless otherwise specified (see 6.2), the heater shall operate at 150 lbf/in^2 pressure and shall withstand a hydrostatic test of 225 lbf/in^2 (see 4.7).

3.4.2 Insulation resistance. Hot and cold insulation resistance tests shall be performed after all welding and allied processes have been completed. The insulation resistance tests as specified in 4.8 shall be in accordance with MIL-H-22577 and UL 574.

3.4.3 Performance for type B heaters. Type B heaters shall perform in accordance with specification requirements when tested as specified in 4.9. This test shall include all flow rates and inlet temperatures described in 4.9.

3.4.4 Vibration. Heaters shall withstand the environmental vibration test specified in 4.10 without mechanical damage or malfunctioning.

3.4.5 Shock. Heaters shall withstand high-impact shock as specified in 4.11 for grade A, class I equipment in accordance with MIL-DTL-901.

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3.4.6 Dielectric strength. Heating elements shall withstand the dielectric test specified in 4.12.

3.5 Identification and information plates. Each heater shall have an identification plate in accordance with types A, B, or C of MIL-DTL-15024 mounted on the terminal box or on the heater lagging. The plate shall contain the following information:

a. Fluid to be heated.

b. Capacity in gallons per minute (type B only).

c. Serial number and type of heater.

d. Electrical characteristics: voltage, wattage, number of phases, frequency, and number of elements.

e. National Stock Number (if available).

f. Name and address of manufacturer.

g. Manufacturer's part number.

3.6 Workmanship. Heaters shall be free from defects affecting their appearance or defects that may affect their operation. Mating surfaces of flanges and mounting pads shall be machined true to provide interchangeability.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.1.1 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions of MIL-STD-202.

4.2 Qualification inspection. Qualification inspection shall be performed on one heater of each type and capacity offered for inclusion on the qualified products list. A qualification inspection shall also be performed when any change in design affects performance characteristics. The heater shall be subjected to the qualification tests specified in table VI (see 6.3).

TABLE VI. Examination and tests.

Examination or test	Requirement	Test method	Qualification	Conformance
Examination (all)	3.2, 3.3, 3.5, 3.6	4.4	X	X
Examination (type A)	3.3.14	4.4	X	X
Examination (type B)	3.3.15 (less 3.3.15.8)	4.4	X	X
Calibration (all)	3.3.5.1 and 3.3.13	4.5	X	X
Reliability and maintainability	3.3.7	4.6	X	
Hydrostatic (all)	3.4.1	4.7	X	
Hydrostatic (type B)	3.3.15.8	4.7	X	
Insulation resistance (all)	3.4.2	4.8	X	X
Performance (type B)	3.4.3	4.9	X	X
Vibration (all)	3.4.4	4.10	X	
Shock (all)	3.4.5	4.11	X	
Dielectric (all)	3.4.6	4.12	X	X
Cybersecurity (all)	3.3.6.3.h	4.13	X	

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4.3 Conformance inspection. Each heater shall be subjected to the conformance inspections specified in table VI. If conformance inspection reveals variations beyond the normal manufacturing tolerance, a complete qualification inspection shall be made on that particular type or size of heater to demonstrate that it conforms to this specification.

4.4 Examination. Each heater offered for delivery shall be examined to ascertain that it is in accordance with this specification. Any heater in the sample containing one or more defects shall be cause for rejection (see table VII).

TABLE VII. Classification of defects.

Categories	Defects
Critical:	
1	Overheating of oil or heating elements at any point within the heater
2	Cybersecurity measures ineffective
Major:	
101	Size not as specified
102	Incomplete, component parts missing (heating elements, over-temperature safety, temperature control, float switch as applicable)
103	Materials defective or not as specified
104	Limiting dimensions exceeded
105	Flange and mounting holes not as specified
106	Welding incomplete, not free of cracks, nonfusion, heavy porosity, heavy undercut, slag inclusions
107	Corner of flanges not smooth, square, or grooved (not pressure tight)
108	Access cover plates missing
109	Gaskets missing or defective
110	Splices in wire
111	Electrical connections not at terminal of component or terminal blocks
112	Wiring not neat
113	Bolts, nuts, and screws not tight, missing, or not as specified
114	Sharp edges and burrs not removed from parts
115	Finish not conforming as specified
116	Marking, identification plate not complete, missing, not permanent, illegible, or not as specified
117	Creepage and clearance distances not as specified

4.5 Calibration. The calibration of the temperature control devices and the over-temperature safety devices shall be checked by a controlled temperature liquid bath system or by use of a test thermocouple attached to the sensing bulb or thermowell of the sensing device to indicate the temperature at which the device functions.

4.6 Reliability and maintainability.

4.6.1 Reliability. The heater shall meet the requirements for reliability as specified in 3.3.7. The validation procedures in MIL-HDBK-217 may be used.

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4.6.2 Maintainability. The heater shall meet the requirements for maintenance and repair as specified in 3.3.7. The maintenance procedures, removal of parts for repair, and the time required for accomplishment shall be verified during conformance testing. The demonstration procedures in MIL-HDBK-470 may be used.

4.7 Hydrostatic test. The heater shall be tested at the specified hydrostatic pressure (see 3.4.1 and 6.2). The expected working fluid (oil or fuel) may be used instead of water for this testing.

4.8 Insulation resistance test. This resistance test shall be as specified in 4.8.1 and 4.8.2 (see 3.4.2).

4.8.1 Insulation resistance, cold. The insulation resistance shall be measured with 500 VDC at room temperature to determine conformance to 3.4.2.

4.8.2 Insulation resistance, hot. The insulation resistance shall be measured with a minimum of 500 VDC within 5 seconds after operation of the element at maximum operating temperature to determine conformance to 3.4.2. To prevent the element from being cooled before the test, the ambient fluid shall not be circulated after the power is removed.

4.9 Performance test for type B heaters. Type B heaters and associated controls shall be tested as specified in 4.9.1 and 4.9.2.

4.9.1 Capacity test. With inlet fluid temperatures and flow rates as specified in table VIII, the following shall be determined:

a. Outlet temperature.

b. Volts and amperes.

c. Pressure drop from inlet to outlet of heater.

d. Number and location of heater elements energized.

e. Thermocouple readings (sufficient thermocouples shall be installed to indicate temperatures in any areas assessed to potentially be hot spots within the heater caused by insufficient flow).

All tests shall be conducted using lubricating oil in accordance with MIL-PRF-17331 (military symbol 2190 TEP).

4.9.2 Pressure drop. The pressure drop shall be determined from the heater inlet to outlet for the flow rates specified in table VIII with the heater operating at rated input power and the specified minimum inlet fluid temperature.

TABLE VIII. Temperatures and flow rates.

Flow (gallons per minute)	Inlet temperatures (°F)
25% capacity	40
	60
	90
50% capacity	40
	60
	90
75% capacity	40
	60
	90
100% capacity	40
	60
	90

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4.9.3 Venting. When specified (see 6.2), the ability of the heater to be drained, filled, and manually vented shall be demonstrated. The demonstration shall include inclined operation in accordance with MIL-DTL-917 (15 degrees in any direction from normal operating position for surface ship heaters and 30 degrees for submarine heaters). Each run shall be for a 1-hour period at rated flow with inlet oil at 90 °F temperature and 20 lbf/in^2 gage pressure. During the test, air shall be introduced into the inlet. The vent connection shall be opened periodically and checked for presence of air to ensure that the heater is venting properly. At the completion of the test, the elements shall be visually examined for damage and traces of carbon deposits possibly caused by the existence of an unvented air pocket.

4.9.4 Temperature control (lubricating oil only). With the type B heater completely filled with oil and not circulating, full power shall be applied for 8 hours. The thermostatic switch shall be set to cut in at 155 °F, decreasing temperature and set to cut out at 160 °F, increasing temperature. Areas of possible hot spots shall be monitored by installed thermocouples and temperatures recorded at 1-hour intervals. At the conclusion of the test, a sample of oil shall be subjected to the following tests:

a. Precipitation number (maximum 0.01) in accordance with FED-STD-791.

b. Neutralization number in accordance with ASTM D974.

4.10 Vibration test. The heater shall be subjected to the type I vibration test in accordance with MIL-STD-167-1.

4.11 Shock test. The heater shall be subjected to the type A, high-impact shock test in accordance with MIL-DTL-901 for grade A, class I equipment.

4.12 Dielectric test. The dielectric test shall be as specified in 4.12.1 through 4.12.3.

4.12.1 General. The dielectric test shall be made after all other tests have been completed. The test voltage shall be equal to twice the rated value of the terminal voltage plus 1,000 volts. The frequency shall be not less than 60 hertz and the waveshape shall approximate a sine wave. The test voltage shall be applied continuously for a period of not less than 1 minute.

4.12.2 Measurements of test voltage. The measurements of the voltage used in the dielectric test shall be made by means of a voltmeter that shall derive its voltage directly from the high voltage circuit, either by means of a voltmeter coil placed in the testing transformer or through an auxiliary potential transformer.

4.12.3 Points of application. The test voltage shall be successively applied between each electric circuit and the frame with all other electric circuits and metal parts grounded.

4.13 Cybersecurity measures verification. All cybersecurity measures required herein (see 3.3.3, 3.3.5, 3.3.5.4, 3.3.5.5, and 3.3.6.3) shall be in accordance with CNSSI No. 1253. Additional cybersecurity verifications may be as specified (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Type A and B heaters are intended for use on naval vessels, meeting shock and other specified requirements in excess of usual commercial design, in the following applications:

a. Type A heaters. Type A heaters are intended for installation in machinery lubricating oil sump tanks to maintain the proper temperature and viscosity of the oil where little circulation or heat is provided (for example, reduction gear sumps).

b. Type B heaters. Type B heaters are intended for use in lubricating oil and fuel systems where heat is required in purifying or to raise the temperature to operating limits required for the applicable machinery.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and style of heater (see 1.2).

c. Operating fluid (see 3.1.1).

d. Materials required (see 3.2).

e. Sheath material for heater elements if other than specified in table IV (see 3.3.2).

f. Terminal box and enclosure(s) locking requirements (see 3.3.3).

g. Terminal box hole size and location (see 3.3.3).

h. Size, type, and surface finish of mounting flange (see 3.3.4 and 3.3.14.1.a).

i. Temperature sensing devices required (see 3.3.5, 3.3.5.4, and 3.3.5.5).

j. Heater control device, specified operating temperature range, and safe default setting (see 3.3.5 and 3.3.6).

k. Heater control required, power requirements, and specific degree of heater control required (see 3.3.6.3.a through 3.3.6.3.f).

l. Additional cybersecurity measures and verifications, if any (see 3.3.6.3.h and 4.13).

m. Baseline configuration parameters, in addition to those listed in 3.3.6.3.h.(10).

n. Voltage requirements (see 3.3.11).

o. If the control devices are not required to operate on the same voltage as the heating elements (see 3.3.11).

p. Over-temperature safety devices (see 3.3.13.1 and 3.3.13.2).

q. Type A heater characteristics (see 3.3.14.1).

r. Size, type, and (for flanges) surface finish of pipe connections (see 3.3.15.2).

s. Mounting configuration (see 3.3.15.4).

t. Insulation or heat shield (see 3.3.15.5).

u. Capacity (see 3.3.15.7).

v. Pressure drop if other than specified (see 3.3.15.8).

w. Limiting dimensions (see 3.3.15.9).

x. Operating pressure if other than as specified (see 3.4.1).

y. Inspection conditions if other than specified (see 4.1.1).

z. Venting ability if required (see 4.9.3).

aa. Packaging requirements (see 5.1).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24299 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Existing stocks. Heaters procured to this specification are form, fit, and function replacements for prior revisions. Further use of heaters procured to prior revisions of this specification should be discontinued and all future requisitions should be to the current revision of this specification.

6.5 Subject term (key word) listing.

Controller, solid-state

Ferrous

Heater, circulation

Heater, immersion

Nonferrous

Safety device, over-temperature

Supervisory network

Thermistor

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – CQ

Preparing activity:
Navy – SH
(Project 4540-2017-001)

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